ASGHINLABS ACADEMY

ICT

FULL LESSON NOTES

COMPLETE · LESSONS 1–105

LOWER SIXTH

GCE A Level · MINESEC Progression

26 chapters · every lesson of the official MINESEC progression sheet, with Python, diagrams and worked examples

ASGHINLABS

L6

LEARN · PRACTISE · SUCCEED

By **Mr Ayemelong S. Ghislain**

www.asghinlabs.com

Academic Year **2026–2027**

ASGHINLABS ACADEMY

Lesson notes · Workbooks · Interactive lessons · AI Tutor

AsghinLabs Academy helps Cameroonian students prepare for the GCE with study materials that follow the official MINESEC progression. These notes are one part of the collection. Everything below is available from our website.

Workbooks available

Subject	What is available
Mathematics	Form 1, Form 2 and Form 3 workbooks · Complete O Level pack · Complete A Level pack (Lower & Upper Sixth)
Computer Science	Form 1, Form 2 and Form 3 workbooks · Complete O Level pack · Complete A Level pack (Lower & Upper Sixth)
ICT	Complete A Level pack (ICT 0796) with GCE past questions

How to order: pick your subject and class on the website, tap "Add to order", then send the order on WhatsApp. No account is needed and no payment is taken on the website. We reply to confirm the total, payment and delivery. Schools and classes ordering 10 copies or more can ask for a group price.

Order online: www.asghinlabs.com/academy.html#workbooks

Order on WhatsApp: +237 682 402 876

ASGHINLABS AI TUTOR

Stuck on an example or a practice question in these notes? Ask the AsghinLabs AI Tutor to explain it step by step, at any time of day.

Open the AI Tutor: www.asghinlabs.com/tutor.html

More from AsghinLabs Academy

- **Full lesson notes:** Lower and Upper Sixth Mathematics, Pure Mathematics with Mechanics, Biology, Physics, Chemistry, Computer Science and ICT, aligned with the MINESEC progression
- **Free interactive lessons:** www.asghinlabs.com/lessons.html
- **Free notes and resources:** www.asghinlabs.com/academy.html#resources
- **Private classes and GCE exam preparation:** www.asghinlabs.com/academy.html
- **WhatsApp channel (updates and new materials):** Join the AsghinLabs channel

www.asghinlabs.com · WhatsApp +237 682 402 876 · asghinlabs@gmail.com

Table of Contents

How to use these notes 9

Chapter 1: Computing Systems and Components 10

Lesson 1: History and Evolution of Computing 10
Early calculating devices · Babbage and Lovelace · five generations of computers · Moore's law · current trends

Review Exercise: Computing Systems and Components 12
Mixed exam-style questions with answers covering the whole topic

Chapter 2: Exploring AI Concepts 13

Lesson 2: Introduction to Artificial Intelligence (AI) 13
Definition of AI · why AI is practical now · narrow, general and super AI · reactive and limited-memory AI · everyday uses

Lesson 3: AI Ethics and Responsible Use 15
Principles of responsible AI · algorithmic bias · deepfakes and misinformation · responsible use by students

Lesson 4: AI Techniques and Intelligent Systems 16
Expert systems · ML, neural networks, NLP, computer vision, fuzzy logic · forward and backward chaining · intelligent agents

Lesson 5: Machine Learning 17
ML vs programming · supervised, unsupervised, reinforcement learning · training and test sets · overfitting · accuracy

Lesson 6: Developing AI Systems 18
Stages of developing an AI system · data quality and GIGO · data split · tools

Review Exercise: Exploring AI Concepts 19
Mixed exam-style questions with answers covering the whole topic

Chapter 3: Description of computing trends and categorization of computers 20

Lesson 7: Types of Computers 20
Supercomputers, mainframes, minicomputers, microcomputers, embedded · analogue, digital, hybrid · general and special purpose

Lesson 8: Basic Components of a Computer 22
Hardware, software, data, people · the IPOS cycle · main components · control unit, ALU, registers

Lesson 9: Input devices 24
Keyboard, mouse, touch screen, scanner, microphone, camera · barcode, OMR, OCR, MICR, biometrics, sensors

Review Exercise: Description of computing trends and categorization of computers 26
Mixed exam-style questions with answers covering the whole topic

Chapter 4: Hardware and Categorisation of Hardware components 27

Lesson 10: Output devices 27
Monitors, projectors, printers, plotters, 3D printers, speakers, actuators · softcopy and hardcopy · inkjet vs laser

Lesson 11: Secondary Storage media and devices 29
Magnetic, optical, solid-state and cloud storage · HDD vs SSD · sequential and direct access

Lesson 12: Primary Storage devices 31
RAM and ROM · DRAM, SRAM, PROM, EPROM, EEPROM · cache memory and registers · virtual memory

Lesson 13: Processing device and the machine instruction cycle 33
CPU parts and registers · the fetch–decode–execute cycle · factors affecting CPU speed

Lesson 14: Processor architectures, parallel & distributed computing 34
Von Neumann and Harvard · CISC and RISC · multi-core · parallel and distributed computing

Lesson 15: Conversion between units of storage and units of processing 35
Bits, bytes, KB, MB, GB, TB · conversions · image file size · Hz, MIPS, FLOPS, bps · download time

Review Exercise: Hardware and Categorisation of Hardware components 37
Mixed exam-style questions with answers covering the whole topic

Chapter 5: Computer Software 38

Lesson 16: Definitions and classification of software 38
Software and programs · system vs application software · proprietary, open source, freeware, shareware · licences

Lesson 17: Application software 40
General-purpose, special-purpose, bespoke, integrated suites · off-the-shelf vs bespoke · generic applications

Review Exercise: Computer Software 41
Mixed exam-style questions with answers covering the whole topic

Chapter 6: Operating Systems 42

Lesson 18: System software and examples 42
Types of system software · utilities and their uses · drivers, translators, firmware

Lesson 19: Notions of the Operating System 44
Definition of the OS · kernel and shell · the boot process · types of operating systems

Lesson 20: Functions of an operating system 1 46
Processes and scheduling · multitasking · deadlock · memory allocation and protection · virtual memory and thrashing

Lesson 21: Functions of the operating system 2 47
File management · device management · spooling and interrupts · security, user interface, error handling

Lesson 22: Installing an operating system and user interfaces 48
Steps to install an OS · partitioning and formatting · file systems · CLI, GUI, menu, touch and voice interfaces

Lesson 23: Using the GUI of an operating system 49
Desktop, icons, windows, menus, taskbar · managing files and folders · keyboard shortcuts

Lesson 24: Using the CLI of an operating system 50
The command prompt · Windows and Linux commands · wildcards · absolute and relative paths

Review Exercise: Operating Systems 52
Mixed exam-style questions with answers covering the whole topic

Chapter 7: Hardware and Software Maintenance 53

Lesson 25: Hardware faults identification and correction 53
Common hardware faults and fixes · troubleshooting steps · preventive maintenance and safety

Lesson 26: Software faults identification and correction 55
Slow PC, frozen programs, BSOD, malware, driver problems · Task Manager, updates, System Restore

Review Exercise: Hardware and Software Maintenance 56
Mixed exam-style questions with answers covering the whole topic

Chapter 8: Assistive technologies and computer ergonomics 57

Lesson 27: Assistive technology 57
Assistive technologies for visual, hearing, motor and learning impairments · accessibility

Lesson 28: Computer ergonomics 59
Ergonomics · RSI, back pain, eye strain · electrical and fire hazards · correct sitting position

Review Exercise: Assistive technologies and computer ergonomics 60
Mixed exam-style questions with answers covering the whole topic

Chapter 9: Using a word processor 61

Lesson 29: Editing and Formatting text 61
Selecting, copying, moving · find and replace · character and paragraph formatting · styles · spelling check

Lesson 30: Editing and formatting images and tables 63
Inserting and formatting pictures · text wrapping · captions · creating and formatting tables · table formulas

Lesson 31: Using Text boxes and adjusting Page layout 64
Text boxes · margins, orientation, size, columns, breaks · headers, footers, page numbers · mail merge

Review Exercise: Using a word processor 65
Mixed exam-style questions with answers covering the whole topic

Chapter 10: Using electronic spreadsheets 66

Lesson 32: Introduction to spreadsheets 66
Cells, rows, columns, ranges · labels, values, formulas · AutoFill · formatting, sorting and filtering

Lesson 33: Performing calculations using spreadsheets 68
Operators and BODMAS · SUM, AVERAGE, MAX, MIN, COUNT, IF, COUNTIF, RANK · error values

Review Exercise: Using electronic spreadsheets 70
Mixed exam-style questions with answers covering the whole topic

Chapter 11: Digital Citizenship and Ethics 71

Lesson 34: Types of cells referencing and calculations 71
Relative, absolute and mixed references · the F4 key · percentages of a total · references to other sheets

Lesson 35: Positive and Negative Uses of Computer Systems 73
Positive uses in education, health, business, communication · negative effects and misuse

Lesson 36: Computer Ethics, Legislation and Cameroon Law 74
Computer ethics · codes of conduct · Cameroon laws on cybersecurity, communications, data protection and copyright · ANTIC

Lesson 37: Data Protection, Copyright and the Digital Divide 76
Personal data and its protection · copyright, plagiarism, patents, trademarks · the digital divide

Review Exercise: Digital Citizenship and Ethics 78
Mixed exam-style questions with answers covering the whole topic

Chapter 12: Cybersecurity 79

Lesson 38: Protecting Computer Systems from Illegal Access 79
Passwords and authentication · biometrics · access rights · firewalls · encryption · physical security

Lesson 39: System Recovery and Safe Working Practices 81
Backups (full, incremental, differential) · System Restore, disk images, UPS, RAID · 3-2-1 rule · safe practices

Lesson 40: Computer Crimes and Combat Measures 82
Hacking, phishing, identity theft, scams, cyberbullying, piracy, DoS · technical, legal and awareness measures

Lesson 41: Malware — Types and Characteristics 83
Viruses, worms, Trojans, ransomware, spyware, keyloggers, adware, rootkits, botnets · signs and spread of infection

Lesson 42: Protecting a Computer System from Malware 84
Antivirus, updates, firewall · safe downloads and e-mail · scanning removable media · backups · user education

Review Exercise: Cybersecurity 85
Mixed exam-style questions with answers covering the whole topic

Chapter 13: Description of Systems 86

Lesson 43: Types of Systems 86
Definition and elements of a system · natural, man-made, open, closed, deterministic, probabilistic · feedback

Lesson 44: Modelling a System with a Data Flow Diagram 88
DFD symbols · context diagram and level 1 DFD · rules for drawing DFDs

Review Exercise: Description of Systems 90
Mixed exam-style questions with answers covering the whole topic

Chapter 14: Information Systems and Data processing 91

Lesson 45: Introduction to Information Systems 91
Data and information · components of an IS · qualities of good information · functions of an IS

Lesson 46: Types of Information Systems 92
TPS, MIS, DSS, EIS, expert systems, office automation, GIS · organisational levels

Lesson 47: General and Commercial Data Processing 93
The data processing cycle · batch, online, real-time, distributed processing · validation checks · master and transaction files

Review Exercise: Information Systems and Data processing 95
Mixed exam-style questions with answers covering the whole topic

Chapter 15: Artificial Intelligence and emerging technologies 96

Lesson 48: Computer Systems in Industry and Science 96
CAD/CAM, process control, stock control · data logging · science, weather, medicine and agriculture

Lesson 49: Robots and Their Application in Daily Life 98
Sensors, controllers, actuators · industrial, service, medical, exploration and agricultural robots · advantages and disadvantages

Lesson 50: Monitoring systems, Control systems, and Monitoring & Control systems 99
Monitoring, control, and monitoring & control systems · sensors · ADC and DAC · feedback · greenhouse algorithm

Lesson 51: Simulation and Modelling 101
Models and simulations · flight, weather, traffic, financial and epidemic models · advantages and disadvantages

Lesson 52: Embedded systems and IoT 102
Embedded systems and their features · Internet of Things applications · benefits and risks

Lesson 53: Virtual Reality and Augmented Reality 104
Virtual reality, augmented reality, mixed reality · equipment · applications · disadvantages

Lesson 54: Multimedia Systems and Authoring 105
Elements of multimedia · file formats · lossy and lossless compression · authoring tools · storyboards

Lesson 55: Multimodal Systems 106
Multimodal interaction · multimodal AI · examples · advantages and disadvantages

Lesson 56: AI technologies 107
NLP, speech, computer vision, recommendation, expert systems, machine and deep learning, robotics · neural networks

Lesson 57: Generative AI 109
Generative AI and LLMs · text, image, audio, video and code generators · prompts · benefits and risks

Lesson 58: Using generative AI for day-to-day activities 110
Generative AI for studying, writing, translation, business, programming · rules for responsible daily use

Review Exercise: Artificial Intelligence and emerging technologies 111
Mixed exam-style questions with answers covering the whole topic

Chapter 16: SDLC and SDLC models 112

Lesson 59: Introduction to the System Development Life Cycle (SDLC) 112
SDLC stages and outputs · TELOS feasibility · fact-finding methods

Lesson 60: SDLC Models 114
Waterfall, V-model, incremental, spiral, agile, RAD · advantages and disadvantages · choosing a model

Lesson 61: The prototyping model 115
Prototypes · steps of prototyping · throw-away and evolutionary prototypes · advantages and disadvantages

Lesson 62: Changeover Strategies 116
Direct, parallel, pilot and phased changeover · data conversion, training and documentation

Review Exercise: SDLC and SDLC models 117
Mixed exam-style questions with answers covering the whole topic

Chapter 17: Modelling Data in an Information System 118

Lesson 63: Data Modelling Concepts 118
Database and DBMS · entities, attributes, records, tables · primary, foreign and composite keys · flat files vs relational · levels of data model

Lesson 64: Entity Relationship Model Concepts and representation systems 120
ER diagrams · Chen and crow's-foot notation · 1:1, 1:M, M:N relationships · participation

Lesson 65: Designing an ER diagram and normalisation 122
Steps to design an ER diagram · resolving M:N relationships · anomalies · first normal form

Lesson 66: Normalisation and converting ER model to relational model 123
2NF and 3NF · partial and transitive dependencies · converting an ER model into tables

Lesson 67: Implementing a relational model 124
Creating tables in a DBMS · data types and field properties · validation rules · relationships and referential integrity · SQL CREATE TABLE

Lesson 68: Using an RDBMS for sorting, filtering and queries 125
Sorting and filtering · queries and criteria · AND and OR criteria · calculated fields, totals and parameter queries · SQL SELECT

Review Exercise: Modelling Data in an Information System 127
Mixed exam-style questions with answers covering the whole topic

Chapter 18: Software Testing 128

Lesson 69: Types of software testing 128
Unit, integration, system, acceptance testing · black-box and white-box · alpha, beta, regression · verification and validation

Lesson 70: Module testing 130
Module testing · normal, boundary and erroneous data · test plans and test logs · stubs and drivers

Review Exercise: Software Testing 132
Mixed exam-style questions with answers covering the whole topic

Chapter 19: Representing numbers 133

Lesson 71: Introduction to data representation and number systems 133
Why binary · bits and bytes · binary, octal, denary, hexadecimal · place values · ASCII and Unicode · uses of hex

Lesson 72: Conversion between number systems and representing negative numbers 135
Denary to binary · binary to hex and octal · sign and magnitude, one's and two's complement · range

Lesson 73: Subtraction and representing negative numbers 136
Binary addition · subtraction with two's complement · overflow

Review Exercise: Representing numbers 137
Mixed exam-style questions with answers covering the whole topic

Chapter 20: Building Logic Circuits 138

Lesson 74: Logic gates 138
NOT, AND, OR, NAND, NOR, XOR · symbols, expressions and truth tables · universal gates

Lesson 75: Logic circuits 140
Combining gates · truth tables of circuits · the half adder · circuits from problem statements

Lesson 76: Boolean expressions from truth table and De Morgan's laws 142
Sum-of-products expressions from truth tables · De Morgan's laws · proof by truth table

Lesson 77: Simplification of Boolean expressions 143
Laws of Boolean algebra · simplifying expressions · why simplification matters

Review Exercise: Building Logic Circuits 144
Mixed exam-style questions with answers covering the whole topic

Chapter 21: Developing Software — Data Types and Structures 145

Lesson 78: Data Types and Data structures 145
Integer, real, character, string, Boolean, date · variables and constants · arrays, records, stacks, queues, lists, trees

Lesson 79: Operations on data structures 147
Array operations in Python · traverse, search, insert, delete, sort · stack and queue operations

Review Exercise: Developing Software — Data Types and Structures 149
Mixed exam-style questions with answers covering the whole topic

Chapter 22: Designing Software 150

Lesson 80: Approaches to Software Design 150
Top-down, bottom-up, modular, structured and object-oriented design · advantages · cohesion and coupling

Lesson 81: Applying software design techniques 152
Top-down design of a report-card program · inputs, processes and outputs of a mobile-money withdrawal

Lesson 82: Representing Software Design 153
Structure charts, flowcharts, pseudocode, DFDs, decision tables, UML

Review Exercise: Designing Software 154
Mixed exam-style questions with answers covering the whole topic

Chapter 23: Writing and Testing Algorithms 155

Lesson 83: Introduction to Algorithms 155
Definition and properties of algorithms · problem-solving steps · everyday algorithms

Lesson 84: Algorithmic instructions 157
Input, output, assignment, selection, iteration · arithmetic, comparison and logical operators · DIV and MOD

Lesson 85: Representing Algorithms — Flowcharts 158
Flowchart symbols · drawing flowcharts with decisions · loops in flowcharts

Lesson 86: Representing Algorithms — Pseudocode 160
Pseudocode conventions · writing pseudocode · flowcharts vs pseudocode

Lesson 87: Control Structures — Sequence and Selection 161
Sequence and selection · IF, nested IF and ELSE IF · CASE · compound conditions

Lesson 88: Control Structures — Loops 162
FOR, WHILE and REPEAT loops · counters and totals · input validation · sentinel values

Lesson 89: Sorting and Searching 164
Linear search · binary search with a trace · bubble sort · selection and insertion sort

Lesson 90: Establishing Correctness of Algorithms.....166
Dry runs and trace tables · syntax, logic and run-time errors · test data

Lesson 91: Evaluating Algorithm Performance.....168
Time and memory efficiency · Big O notation · linear vs binary search · choosing algorithms

Review Exercise: Writing and Testing Algorithms.....169
Mixed exam-style questions with answers covering the whole topic

Chapter 24: Programming Paradigms and Software Reuse.....170

Lesson 92: Programming Paradigms.....170
Procedural, object-oriented, functional, logic, declarative and event-driven paradigms · OOP concepts · low- and high-level languages

Lesson 93: Software Reuse.....172
Libraries, modules, frameworks, APIs · advantages and disadvantages of reuse

Lesson 94: Internally versus Externally Developed Software.....173
In-house, outsourced, off-the-shelf, open-source and SaaS software · decision factors

Review Exercise: Programming Paradigms and Software Reuse.....174
Mixed exam-style questions with answers covering the whole topic

Chapter 25: Implementing algorithms.....175

Lesson 95: Language Translators and IDE.....175
Assemblers, compilers, interpreters · compiler vs interpreter · features of an IDE

Lesson 96: introduction to coding.....177
Python basics · input, print, conversion · operators · naming rules and indentation

Lesson 97: Writing code 1.....178
Sequence programs: temperature conversion, VAT, time conversion

Lesson 98: Writing code 2.....179
if, elif, else · grading, leap years, taxi fares

Lesson 99: writing code 3.....180
for loops and range · while loops with sentinel values · input validation

Lesson 100: writing code 4.....181
Lists: create, access, append, remove, sort, search, sum, max · processing parallel lists

Lesson 101: writing code 5.....182
String operations · counting vowels · a menu-driven calculator

Review Exercise: Implementing algorithms.....184
Mixed exam-style questions with answers covering the whole topic

Chapter 26: Writing functions and testing a developed program.....185

Lesson 102: Functions and procedures 1.....185
Subroutines, functions and procedures · parameters and arguments · local and global variables · advantages

Lesson 103: Functions and procedures 2.....187
Writing useful functions · grade, average, is_prime · built-in and user-defined functions · passing values

Lesson 104: Testing a program 1.....188
Syntax, run-time and logic errors · test plans · testing a validation function

Lesson 105: Testing a program 2.....189
Debugging techniques · try/except error handling · correcting logic errors · documentation

Review Exercise: Writing functions and testing a developed program.....190
Mixed exam-style questions with answers covering the whole topic

How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

Need help? Ask the AsghinLabs AI Tutor at **www.asghinlabs.com/tutor.html** or message us on WhatsApp (+237 682 402 876).

Chapter 1: Computing Systems and Components

Computers have changed from room-sized calculating machines to the phones in our pockets. This chapter traces that history and the generations of computers.

Lesson	Title (MINESEC progression)
1	History and Evolution of Computing

Lesson 1: History and Evolution of Computing

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the history of computing devices
- ✔ compare the generations of computers

KEY POINTS

A **computer** is an electronic device that accepts data (**input**), processes it according to instructions (a **program**), stores it, and produces results (**output**).

Early calculating devices

Period	Device	Importance
About 3000 BC	Abacus	Beads on rods for counting; still used today
1617	Napier's bones	Rods for multiplication
1642	Pascal's calculator (Pascaline)	Mechanical adding machine with gears
1804	Jacquard loom	Punched cards controlled weaving patterns: the idea of a stored program
1830s	Babbage's Analytical Engine	Design with input, store (memory), mill (processor) and output: the first general-purpose computer design. Babbage is called the "father of the computer"
1843	Ada Lovelace	Wrote an algorithm for the Analytical Engine: the first programmer
1890	Hollerith's tabulator	Punched cards processed the US census; his company became IBM
1936	Alan Turing	The Turing machine: a theoretical model of every computer

Generations of computers

Generation	Technology	Features	Examples
First (1940–1956)	Vacuum tubes	Huge, very hot, unreliable; machine language; punched cards	ENIAC, UNIVAC I
Second (1956–1963)	Transistors	Smaller, faster, more reliable; assembly language and early high-level languages (FORTRAN, COBOL)	IBM 1401
Third (1964–1971)	Integrated circuits (ICs)	Many transistors on a chip; keyboards, monitors, operating systems, multiprogramming	IBM System/360
Fourth (1971–today)	Microprocessors (VLSI)	Whole CPU on one chip; personal computers, GUIs, networks, the internet	Intel 4004, IBM PC, Apple Macintosh
Fifth (today and future)	AI, parallel processing, quantum computing	Natural language, machine learning, robotics	AI assistants, self-driving cars

KEY POINTS

Over the generations computers became **smaller, faster, cheaper, more reliable and more energy-efficient**, with more memory and easier to use.

EXAM TIP

Moore's law (1965): the number of transistors on a chip doubles about every two years.

Computing trends today

- **Mobile computing:** smartphones and tablets.
- **Cloud computing:** storage and software over the internet.
- **Internet of Things (IoT):** everyday objects connected to the internet.
- **Artificial intelligence** and **quantum computing**.

Example: Give two differences between first-generation and fourth-generation computers.

First-generation computers used vacuum tubes, filled whole rooms and used a lot of power; fourth-generation computers use microprocessors, fit on a desk or in a pocket, are much faster and use little power.

Practice Questions: Lesson 1

1. Which technology was used by second-generation computers?
2. Who is called the first programmer?

Answers: *1. Transistors 2. Ada Lovelace*

Review Exercise: Computing Systems and Components

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. Define a computer and state its four basic operations.
2. Explain the contribution of Charles Babbage and of Ada Lovelace.
3. Copy and complete a table of the five generations of computers with the technology used by each.
4. State four ways in which computers have changed from the first to the fourth generation.
5. State Moore's law.

Answers: *1. An electronic device that accepts data, processes it under the control of a program and produces output; input, processing, storage, output 2. Babbage designed the Analytical Engine with input, store, mill and output; Lovelace wrote the first algorithm for it (first programmer) 3. 1st vacuum tubes; 2nd transistors; 3rd integrated circuits; 4th microprocessors; 5th AI and parallel/quantum processing 4. Smaller, faster, cheaper, more reliable (also less power, more memory) 5. The number of transistors on a chip doubles about every two years*

Chapter 2: Exploring AI Concepts

Artificial intelligence is changing work and daily life in Cameroon and around the world. This chapter introduces AI, its ethics, its main techniques, machine learning and how AI systems are developed.

Lesson	Title (MINESEC progression)
2	Introduction to Artificial Intelligence (AI)
3	AI Ethics and Responsible Use
4	AI Techniques and Intelligent Systems
5	Machine Learning
6	Developing AI Systems

Lesson 2: Introduction to Artificial Intelligence (AI)

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define artificial intelligence
- ✔ classify AI and identify its everyday uses

KEY POINTS

Artificial intelligence (AI) is the ability of a computer system to perform tasks that normally need human intelligence: learning, reasoning, solving problems, understanding language and recognising images or speech.

Why AI is practical now

- Cheaper and more powerful processors (including GPUs).
- Huge amounts of digital data (big data) from phones, the web and sensors.
- Better algorithms, especially deep learning.

Types of AI

By capability	Meaning	Exists today?
Narrow (weak) AI	Does one specific task well	Yes: all current AI
General (strong) AI	Would match humans at any intellectual task	No
Super AI	Would be more intelligent than humans	No (theoretical)

By function	Meaning	Example
Reactive machines	React to the present only, no memory	Chess program Deep Blue
Limited memory	Use recent past data to decide	Self-driving cars
Theory of mind	Would understand emotions and beliefs	Research only
Self-aware	Would have consciousness	Hypothetical

AI in everyday life

Area	Example
Communication	Voice assistants, automatic translation, chatbots
Phones	Face unlock, predictive text, photo enhancement
Health	Reading X-rays, predicting disease outbreaks
Agriculture	Detecting crop diseases from leaf photos
Banking and mobile money	Fraud detection, credit scoring
Transport	Route planning, self-driving cars
Education	AI tutors, automatic marking, personalised learning

Example: Classify a face-unlock system by capability and by function.

Narrow AI (one task) and **limited memory** (it uses stored data about the owner's face).

Practice Questions: Lesson 2

1. Give two everyday examples of AI.
2. Which type of AI exists today?

Answers: *1. E.g. voice assistants, face unlock, spam filters 2. Narrow (weak) AI*

Lesson 3: AI Ethics and Responsible Use

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ explain the principles of responsible AI
- ✔ identify risks and misuse of AI

Because AI now helps decide who gets loans, jobs or medical treatment, it must be used **responsibly**.

Principle	Meaning
Fairness	The system must not discriminate against any group
Transparency	People should know when AI is used and why a decision was made
Privacy	Collect only necessary data, with consent, and protect it
Accountability	A human or organisation stays responsible for the AI's decisions
Safety and security	The system must be reliable and protected from misuse
Human oversight	Important decisions can be checked and overridden by people

KEY POINTS

Algorithmic bias gives unfair results for some groups, usually because the **training data** under-represents them or reflects past discrimination.

Risks and misuse

- **Deepfakes:** realistic fake videos or voices used for fraud and false news.
- **Misinformation:** AI can produce convincing false text; AI tools may "hallucinate" (invent facts).
- **Privacy and surveillance:** face recognition can track people.
- **Jobs:** some tasks are automated, while new jobs appear.
- **Copyright:** AI trained on other people's work raises ownership questions.

Using AI responsibly as a student

- Use AI to understand and check, not to copy; verify facts in reliable sources.
- Never enter personal or confidential data into AI tools.
- Follow your school's and the exam board's rules on AI use.

Example: A recruitment app trained on past hiring data rejects most female applicants. Identify the problem and two solutions.

> **Bias** from unbalanced training data. Solutions: retrain on balanced data and test results for each group; let a human review the AI's decisions.

Practice Questions: Lesson 3

1. What is a deepfake?
2. State one rule for using AI responsibly in your studies.

Answers: *1. A realistic fake video, image or voice made with AI 2. E.g. verify facts in reliable sources; do not copy AI answers as your own*

Lesson 4: AI Techniques and Intelligent Systems

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the main AI techniques
- ✔ explain expert systems and intelligent agents

Lesson notes

Technique	How it works	Example
Expert systems	A knowledge base of facts and IF–THEN rules plus an inference engine that applies them	Medical diagnosis, loan advice
Machine learning	Learns patterns from data	Spam filters
Neural networks / deep learning	Layers of artificial neurons with weights learned from data	Face and speech recognition
Natural language processing (NLP)	Understands and produces human language	Chatbots, translation
Computer vision	Interprets images and video	Number-plate readers
Fuzzy logic	Works with degrees of truth between 0 and 1	Washing machines, air conditioners
Search and planning	Explores possible moves or routes to find the best	GPS route finding, game playing
Robotics	Machines that sense and act in the physical world	Factory robots

KEY POINTS

An **expert system** has a **knowledge base**, an **inference engine**, a **user interface** and an **explanation facility**. **Forward chaining** reasons from facts to conclusions; **backward chaining** starts from a goal and looks for supporting facts.

An **intelligent agent** perceives its environment through **sensors** and acts on it through **actuators** to reach a goal.

Worked examples

Example: Identify the sensors and actuators of a smart irrigation system.

Sensors: soil-moisture and temperature sensors, rain sensor. Actuators: pump and valves that turn the water on and off.

Practice Questions: Lesson 4

1. Name the part of an expert system that applies the rules.
2. Which AI technique lets a chatbot understand questions?

Answers: *1. The inference engine 2. Natural language processing (NLP)*

Lesson 5: Machine Learning

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the types of machine learning
- ✔ calculate the accuracy of a model

Lesson notes

KEY POINTS

Machine learning (ML) lets computers learn from **data** instead of being programmed with every rule. Training produces a **model** that makes predictions on new data.

Type	Data	Example
Supervised learning	Labelled examples (inputs with correct answers)	Predicting exam scores; spam detection
Unsupervised learning	Unlabelled data; the model finds groups or patterns	Grouping customers by buying habits
Reinforcement learning	Rewards and penalties from trial and error	Game-playing agents, robot walking

KEY POINTS

Training set: data used to learn. **Test set:** unseen data used to check performance. **Overfitting:** the model memorises the training data and fails on new data.

Accuracy = correct predictions ÷ total predictions × 100%.

Worked examples

Example: A model classifies 186 of 200 test photos correctly. Find its accuracy.

186 ÷ 200 × 100 = **93%**

Task type	Output	Example
Classification	A category	"Ripe" or "unripe" fruit
Regression	A number	Tomorrow's temperature
Clustering	Groups	Customer segments

Practice Questions: Lesson 5

1. Which type of learning uses labelled data?
2. A model is right 45 times out of 50. Find its accuracy.

Answers: *1. Supervised learning 2. 90%*

Lesson 6: Developing AI Systems

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ list the stages of developing an AI system
- ✔ explain the importance of data quality

Stages

1. **Define the problem** and how success will be measured.
2. **Collect data** (surveys, sensors, databases, open datasets).
3. **Prepare the data:** clean errors and duplicates, label it, split it (e.g. 80% training, 20% testing).
4. **Choose and train a model.**
5. **Evaluate** it on the test data; improve if needed.
6. **Deploy** it in an app, website or device.
7. **Monitor and retrain** as new data arrives.

KEY POINTS

GIGO — Garbage In, Garbage Out: poor, wrong or biased data produces a poor, wrong or biased model. Good data is **sufficient, accurate, relevant, representative and legally obtained**.

Tool	Use
Python with scikit-learn, TensorFlow, PyTorch	Building and training models
Jupyter notebooks, Google Colab	Writing and running AI code
Teachable Machine	Training simple image or sound models without code

Example: A school wants an AI tool that predicts which students risk failing. Suggest the data needed and one ethical precaution.

> Data: past marks, attendance, assignment completion (with results). Precaution: protect students' privacy and let teachers, not the AI alone, decide on support.

Practice Questions: Lesson 6

1. What is usually the percentage split between training and testing data?
2. What does GIGO stand for?

Answers: *1. About 80% training and 20% testing 2. Garbage In, Garbage Out*

Review Exercise: Exploring AI Concepts

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. Define artificial intelligence and give three reasons why it has become practical.
2. Distinguish between narrow AI and general AI. Which exists today?
3. State four principles of responsible AI.
4. Explain how bias can enter an AI system.
5. Name the parts of an expert system and explain forward chaining.
6. Describe supervised, unsupervised and reinforcement learning with one example each.
7. A model makes 340 correct predictions out of 400. Calculate its accuracy.
8. List the stages of developing an AI system and explain GIGO.

Answers: *1. The ability of computers to do tasks needing human intelligence; cheap powerful processors, big data, better algorithms 2. Narrow AI does one task; general AI would do any intellectual task like a human; only narrow AI exists 3. Fairness, transparency, privacy, accountability (also safety, human oversight) 4. Through training data that under-represents some groups or reflects past unfair decisions 5. Knowledge base, inference engine, user interface, explanation facility; forward chaining applies rules to known facts to reach new conclusions 6. Supervised: labelled data (spam filter); unsupervised: finds groups (customer segments); reinforcement: rewards (game agent) 7. 340 ÷ 400 = 85% 8. Define problem, collect data, prepare data, train, evaluate, deploy, monitor; bad input data gives bad results*

END OF FREE SAMPLE

Get the full notes

You have just read the opening of **Lower Sixth ICT**. The complete book has every lesson of the MINESEC progression sheet, in order:

- All 105 lessons, each with objectives, clear notes and KEY POINTS
- Worked examples and practice questions with answers
- Review exercises at the end of every chapter

This book: **3 000 FCFA** · Lower + Upper Sixth: **5 000 FCFA** per subject
Complete 5-subject A Level package: **20 000 FCFA**

Order at asghinlabs.com/academy

or WhatsApp **+237 682 402 876** · PDF sent straight to your phone